Water Footprint of India's E20 Transition — Detailed Research Report

25 October 2026 · Team NeerNexus

India's E20 programme has put real and growing pressure on water, but not in the way the original claim suggested. The pressure has moved from sugarcane to maize, and from the monsoon season into the irrigated rabi and summer seasons, adding an estimated 2–9 bcm of groundwater use a year. That is 1–4% of national extraction, and it does not yet show up as rising groundwater stress in the 10 states studied, except Rajasthan and Madhya Pradesh.

Team: NeerNexus (CM 617): Armaan Valecha (23B0432), Om Anghan (23B0338), Akhil Tripathi (23B0308), Vaishnavi Rajput (23B0382).

Supplementary material: dataset NeerNexus_Stage2_Dataset.xlsx (17 tabs: data, source log, gaps and analysis with formulas) and the Stage 1 and Stage 3 working documents. Drive link: to be added by the team.

This report backs the 2-page submission. Every number traces to a source listed in Section 10 and logged in the dataset.

Our journey in brief

Where we started. A first draft of this project claimed that E20 drives water-intensive crops and drains groundwater. We were not sure the claim held up, so we rebuilt the study from scratch with one rule: every number needs a traceable source. We kept MSP in view from the start, because it pushes farmers toward the same crops and could be the real cause.

Stage 1: what is already known. We collected the policy timeline, the contested water-per-litre figures and the research literature, and listed the data we would need. We found no study that tests the claim state by state with recent data, and that became our research gap (Section 2).

Stage 2: building the dataset. We gathered official crop, groundwater, rainfall, MSP and ethanol data into one spreadsheet. Two gaps appeared at once. Groundwater is assessed only in 2017, 2020 and 2022–2024, not every year, and nobody publishes which crops each distillery buys. So we compared 2017 with 2024 and kept the ethanol-by-crop step national (Section 4).

Stage 3: testing the claim link by link. We split the claim into five testable links (Section 3). The first result changed our focus: maize, not sugarcane, is where ethanol takes a large share of the harvest. The groundwater test then surprised us: in most states where these crops expanded, groundwater stress fell, because wet monsoons refilled aquifers (Section 5).

Stage 4: changing approach. State averages could not reveal ethanol's effect, so we estimated the extra groundwater directly from the added dry-season maize, put a cost on it, and drew policy options from what we found (Sections 6 and 7).

Where we ended up. The claim is partly supported. E20 adds groundwater demand through dry-season maize in a few states, but so far it is small nationally and hidden by good monsoons. MSP explains the growth of rice, not maize (Section 8).

Terms used in this report

Term	Meaning
E20	Petrol blended with 20% ethanol

Term	Meaning
ESY	Ethanol supply year, November to October (ESY 2024-25 = Nov 2024 to Oct 2025)
MSP	Minimum support price: the price at which the government promises to buy a crop
FRP	Fair and remunerative price: the minimum price sugar mills must pay farmers for cane
Procurement	The government actually buying a crop at MSP (mainly rice and wheat)
Kharif, rabi, summer	The monsoon season (mostly rain-fed), the winter season and the pre-monsoon dry season (both mostly irrigated)
Stage of extraction (groundwater stress)	Groundwater pumped in a year as a percentage of the amount replenished; 70–90% is semi-critical, 90–100% critical, above 100% over-exploited
bcm	Billion cubic metres (1 bcm = 1,000 billion litres)
Lakh, crore	1 lakh = 100,000; 1 crore = 10 million
H1–H5	The five links of the claim we tested (Section 3)
S1, S3	Water scenarios: S1 counts all crop water; S3 counts by-products (surplus rice, molasses) as zero. S2 was dropped for lack of irrigation data
P1–P5	The five policy pathways (Section 7)
ρ (rho)	Spearman rank correlation, from −1 to +1; with 10 states it must exceed about 0.65 in size to be significant
C-heavy molasses, 2G ethanol	A by-product of sugar making; ethanol from crop residues
Agencies	UPAg: Ministry of Agriculture statistics portal; DES: Directorate of Economics and Statistics; CACP: Commission for Agricultural Costs and Prices; CGWB / CGWA: Central Ground Water Board / Authority; IMD: India Meteorological Department; PPAC: Petroleum Planning & Analysis Cell; FCI: Food Corporation of India; AIDA: All India Distillers' Association; OMC: oil marketing companies

1. Summary of findings

The claim we set out to test is partly supported: E20 is creating new groundwater demand, but through irrigated maize rather than sugarcane, and on a scale that state-level groundwater statistics do not yet register.

1. **E20 has reshaped maize.** About 19–20% of India's maize went to ethanol in 2023-24 and 29–32% in 2024-25. The rise in maize output since 2016-19 (12.3 Mt) roughly equals the maize used for ethanol in 2024-25 (12.5–13.9 Mt), and India began importing maize. Ethanol takes only 2–4% of rice and 0–10% of cane.
2. **The extra maize is increasingly irrigated.** Rabi and summer maize grew from 27% of output in 2016-17 to 43% in 2024-25, and their area rose 70%. These seasons grow outside the monsoon.
3. **MSP alone does not move crops; procurement does.** Millets got the largest MSP rises (+106% to +160%) yet barely grew; across seven crops the rank correlation between MSP rise and output growth is −0.43. Rice expansion follows MSP-backed procurement, not ethanol.
4. **No state-level link yet between crop expansion and groundwater stress (2017–2024).** Expanding states had wetter monsoons that recharged aquifers. Rajasthan (+10 points) and Madhya Pradesh (+3.6 points) are the exceptions.

5. **New groundwater from irrigated maize: about 2–9 bcm a year** (1–4% of national extraction), concentrated in Bihar, Karnataka, Uttar Pradesh and Telangana.
6. **Its subsidy cost is modest per litre but not trivial:** ₹2–10 per litre of maize ethanol, against about ₹41 of forex saved and about 2 kg of CO_2 avoided per litre.
7. **Summer maize is the sharpest lever:** it draws an estimated 3.6–6.0 bcm of groundwater a year for about 150 crore litres of ethanol (14–15% of 2024-25 supply).

	Hypothesis	Verdict	Confidence
H1	A material, rising share of crops goes to ethanol	Supported for maize; weak for rice; unclear for cane	Medium
H2	These crops grew faster after the ethanol push	Supported for maize; partly for cane; rice grew for other reasons	High
H3	Growth follows ethanol demand more than MSP	Supported for maize; MSP and procurement fit rice; cane mixed with rainfall	Medium
H4	The expansion is depleting groundwater	Not supported at state level so far; new demand estimated at 1–4% of extraction	Medium-low
H5	The water cost is significant	Moderate per litre; concentrated in irrigated seasons and a few states	Low-medium

2. Background and research gap

India raised ethanol blending from 1.53% in ESY 2013-14 to an average of 19.2% in ESY 2024-25, reaching 20% in October 2025, and now aims for 30% by 2030. The feedstock base widened from molasses to cane juice and grains, which ties fuel policy to crop and water choices.

Date	Policy step	Source
May 2018	National Policy on Biofuels allows cane juice, damaged grain and surplus grain	PM India
Jul 2018	Interest subvention for distilleries; MSP reset to at least 1.5 times cost	Lok Sabha AU632, PRS
Nov 2020 – Jan 2021	FCI rice and maize allowed; subvention extended to grain distilleries	Lok Sabha AU632
Jun 2021	NITI Aayog roadmap: 1,016 crore L demand by 2025-26; flags about 2,860 L water per L of sugar ethanol	PRS summary
May 2022	E20 target moved from 2030 to ESY 2025-26	PIB
Jul – Dec 2023	FCI rice halted; cane juice barred, then capped at 1.7 Mt of sugar	Deccan Herald, Argus
Aug 2024	Cane juice and FCI rice allowed again	Deccan Herald
Oct 2025	20% blending first reached in a month; ESY 2024-25 average 19.2%	PPAC via BioEnergy Times

The water concern is real but its numbers are contested.

- A widely cited set of figures, 10,790 / 4,670 / 3,630 L of water per L of ethanol for rice / maize / cane, comes from a Kisan Tak–India Today report, with the rice figure credited to the Food Secretary.
- Industry cites an ICAR-IISR study commissioned by ISMA at about 2,000 L for cane.
- The government stresses that distilleries use only 3–5 L of process water per litre and that only surplus grain is used (Rajya Sabha, Jul 2026).

What the literature covers:

- Groundwater science shows pumping drives depletion in northwest India while rainfall drives variability elsewhere (Asoka et al. 2017; Dangar et al. 2021).
- North India lost about 450 km³ of groundwater in 2002–2021 (Mishra et al. 2024).

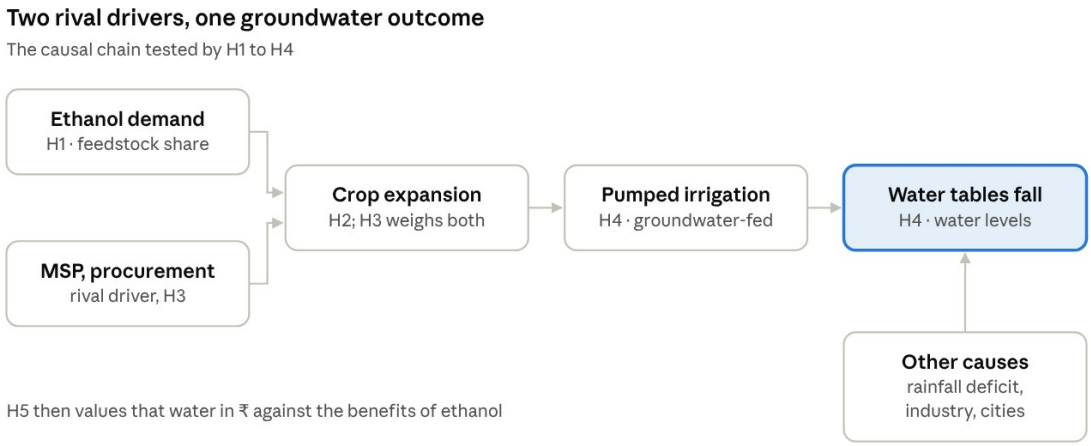
- A national land-use model projects E20 raising agricultural water use 21–118% above a no-mandate path by 2030, depending on feedstock (Jha et al. 2026).

Research gap. No study we found tests, state by state and with post-2018 data, whether ethanol demand actually expanded water-intensive crops and drew down groundwater, or separates that from MSP and procurement. Jha et al. explicitly list regional groundwater stress as future work.

3. Research question and hypotheses

Question. Has India's ethanol push expanded water-intensive crops and drawn down groundwater, and is ethanol demand or price support (MSP or FRP with procurement) the stronger driver of that expansion?

We broke the claim into a chain of testable links, so that each can be confirmed or rejected with data and the overall verdict follows from the parts.



Study design · hypotheses H1–H5

#	Hypothesis	Main test
H1	A material and rising share of cane, rice and maize goes to ethanol	Ethanol volume by feedstock converted to tonnes, as a share of national output
H2	These crops grew faster after the ethanol push	Area and output, 2016-19 average vs 2022-25 average, by state and season
H3	That growth follows ethanol demand more than price support	MSP rise vs output growth across crops; ethanol-only policy shocks; procurement-backed crops
H4	The extra crop is groundwater-irrigated and linked to falling water tables	State change in stage of extraction vs crop expansion, with rainfall and recharge as controls; seasonal irrigation
H5	The water cost is significant relative to ethanol's benefits	Groundwater per litre x pumping subsidy, against forex and CO_2 per litre

The chain decides the verdict. If H1–H4 all hold, the claim stands; if H4 fails, the risk is not yet visible in groundwater; if H3 favours MSP, policy should target procurement before ethanol.

4. Data and methods

We compiled official data series and combined them with simple, transparent calculations (a metadata analysis); no new field data were collected.

Scope.

- **States:** 10 covering each feedstock and the main price-support case. Cane: Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu. Rice and procurement: Punjab, Haryana, Telangana. Maize and grain distilleries: Bihar, Madhya Pradesh, Rajasthan.
- **Period:** 2015-16 to 2024-25, with 2025-26 advance estimates where available. "Before" is 2016-19 (2015-16 to 2017-18 where noted); "after" is 2022-25.
- **Groundwater:** only the CGWB assessment years 2017, 2020, 2022, 2023 and 2024 exist; we compare 2017 with 2024.

Variable	Source	Years	Level
Crop area and production by season (maize, rice, cane, wheat, jowar, bajra, ragi)	UPAg State Profile, DA&FW / DES	2016-17 to 2025-26	State
Earlier crop data (rice, cane, coarse cereals)	RBI Handbook 2023, Tables 60, 69, 71, 80	2014-15 to 2022-23	State
National production	DES time series	2015-16 to 2025-26	National
Groundwater extraction, resource, stage	RBI Table 107, from CGWB	2017–2024	State
Monsoon rainfall	MoES / IMD Parliament answers (2016, 2021, 2023)	2015–2016, 2018–2022	State
MSP and FRP	DES MSP statement; CACP compilation	2015-16 to 2025-26	National
Ethanol volumes, feedstock split, prices	Parliament answers, PPAC, Indian Express, AIDA (Section 10)	2013-14 to 2025-26	National

Calculations.

- **Ethanol share of a crop (H1):** feedstock (t) = ethanol (L) ÷ litres per tonne; share = feedstock ÷ national production. Conversion: maize 380–400 L/t, rice 370–470 L/t, cane juice 70–81 L/t.
- **Growth (H2, H3):** percentage change between three-year averages; MSP rise and output growth over 2015-16 to 2024-25, compared by rank correlation.
- **Groundwater link (H4):** change in stage of extraction (2017 to 2024) against change in rice + sugarcane + rabi/summer maize area (2016-19 vs 2021-24 averages, so the crop window ends with the 2024 assessment), using Spearman rank correlation, with monsoon departure and change in extractable resource as controls.
- **Embedded water (magnitude):** ethanol volume × water per litre. Scenario S1 counts all crop water; S3 counts by-products (surplus FCI rice, C-heavy molasses) as zero.
- **New groundwater (Section 6):** added rabi + summer maize area × irrigation depth (300–700 mm) × groundwater share of irrigation (61–100%).

New groundwater = added rabi and summer maize area × irrigation depth × groundwater share of irrigation

Verification.

- UPAg all-India maize matches the DES time series exactly.
- Sugarcane production rebuilt from area × yield for 2015-16 and 2017-18 matches DES national totals.
- 2017 and 2020 groundwater extraction match a Lok Sabha answer exactly.
- One conflict was found: RBI prints Punjab 2022 as 165.99%, while CGWB's Punjab report gives 164.11%, which matches extraction ÷ resource. We use CGWB.

All calculations are live formulas in the dataset's Analysis tabs, so any input can be changed and the results recompute.

5. Results

Maize is where E20 shows up in the data: a large and rising share goes to ethanol, its output nearly doubled, and its growth is moving into the irrigated seasons. Groundwater statistics at state level do not yet register the effect.

5.1 H1: How much crop goes to ethanol

Why this step: before asking about water, we needed to know which crop ethanol actually draws on. The answer redirected the rest of the study toward maize.

Feedstock	Ethanol 2023-24 (crore L)	Share of output	Ethanol 2024-25 (crore L)	Share of output
Maize	286	19–20%	499–530	29–32%
Rice and broken grain	116	1.8–2.3%	188–219	2.7–3.9%
Sugarcane	270	0–8.5%	321	0–10%

Cane is a range: C-heavy molasses uses no extra cane, so the lower bound is zero; the upper bound assumes every litre came from juice. Grain overtook cane as the main feedstock in this period (60% of supply in 2023-24 and 69% in 2024-25). India also turned to maize imports as distilleries competed with poultry feed (The Hindu via Drishti IAS).

5.2 H2: Which crops grew after the ethanol push

Why this step: if ethanol drives crop choices, the feedstock crops should grow faster after the 2018 push, in particular states and seasons.

Change in area between the 2016-19 and 2022-25 averages (UPAg):

State	Maize area	Rabi + summer maize area (lakh ha)	Rice area	Sugarcane area
Bihar	+26%	4.6 to 6.7	−6%	−14%
Haryana	−18%	0	+10%	−18%
Karnataka	+44%	1.1 to 3.1	+9%	+41%
Madhya Pradesh	+36%	0.2 to 0.8	+61%	+5%
Maharashtra	+37%	3.3 to 4.6	+14%	+47%
Punjab	−19%	0	+5%	−4%
Rajasthan	+8%	0.05 to 0.24	+46%	negligible area
Tamil Nadu	+33%	1.8 to 2.2	+29%	−21%
Telangana	−20%	1.5 to 3.0	+154%	−27%
Uttar Pradesh	+22%	0.4 to 2.4	+8%	+23%
All India	+22%	18.1 to 30.8	+11%	+20%

All-India maize output rose 45% (274.6 to 397.2 lakh t), against 11% for rice area and 20% for cane area. The growth is moving into the irrigated seasons:

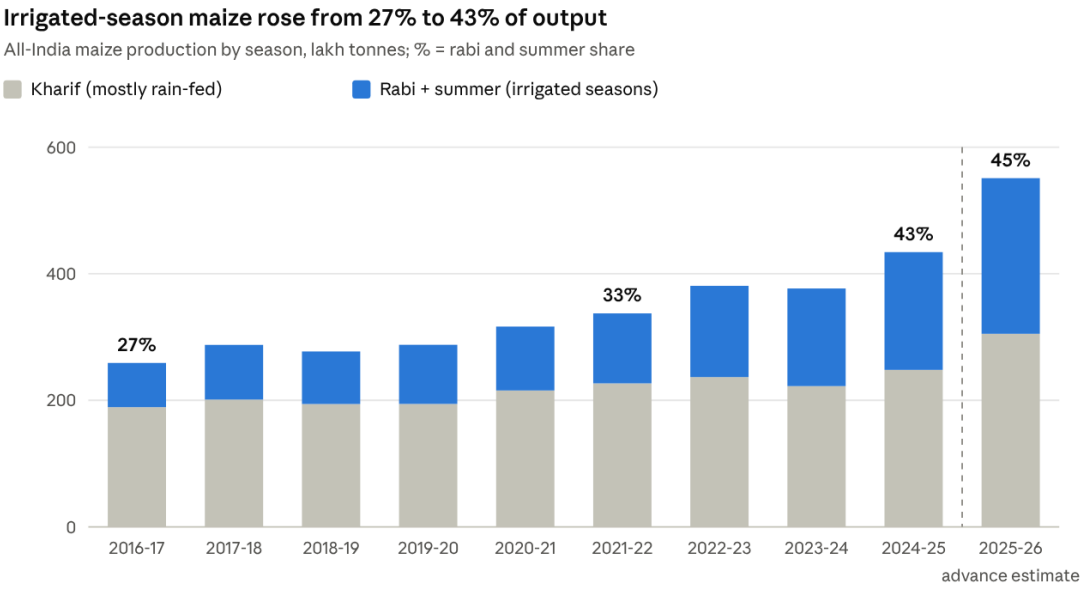


Figure 2 · UPAg State Profile (DA&FW) · All India maize, 2016-17 to 2025-26

Rabi and summer maize area rose 70%, against 10% for kharif. The largest additions were in Bihar, Karnataka, Uttar Pradesh and Telangana. Summer maize has been reported separately only since 2022-23, so we compare rabi and summer together.

5.3 H3: Ethanol or MSP?

Why this step: growth alone does not prove ethanol is the cause, because MSP pushes the same crops. So we compared each crop's MSP rise with its output growth.

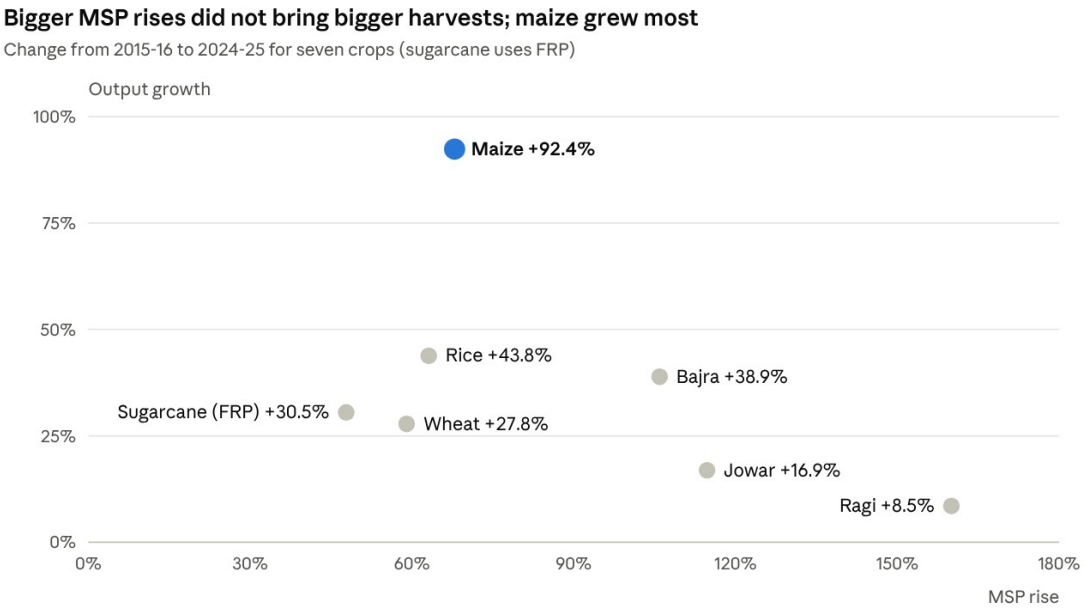


Figure 3 · DES MSP statement, CACP compilation, DES production time series · 2015-16 to 2024-25

- **MSP alone does not explain growth.** Jowar, bajra and ragi got MSP rises of 106–160% but output grew only 8–39%, and jowar area fell 21%. The rank correlation between MSP rise and output growth across the seven crops is −0.43.

- **Maize fits ethanol demand.** Its MSP rose a moderate 68%, but output grew 92%, in step with maize ethanol rising from almost nothing in 2021-22 (0% of supply) to about 500 crore litres in 2024-25. The December 2023 cane-juice curb and the July 2023 FCI rice halt pushed distilleries further toward maize.
- **Rice fits MSP plus procurement.** Rice area grew most in Telangana (+154%), Madhya Pradesh (+61%) and Rajasthan (+46%), while rice for ethanol is only 2–4% of output. In Punjab, MSP and near-total procurement keep paddy the most profitable crop (Sonker & Kallummal, IIFT).
- **Cane is mixed.** Area rose 47% in Maharashtra and 41% in Karnataka, both of which had wetter-than-normal monsoons through 2018–2022 (+16% and +15% on average), so rain and ethanol demand cannot be separated at state level.

5.4 H4: Groundwater against crop expansion

Why this step: this is the heart of the claim. If the extra crops pump more groundwater, stress should rise where they expanded, unless rainfall explains the change.

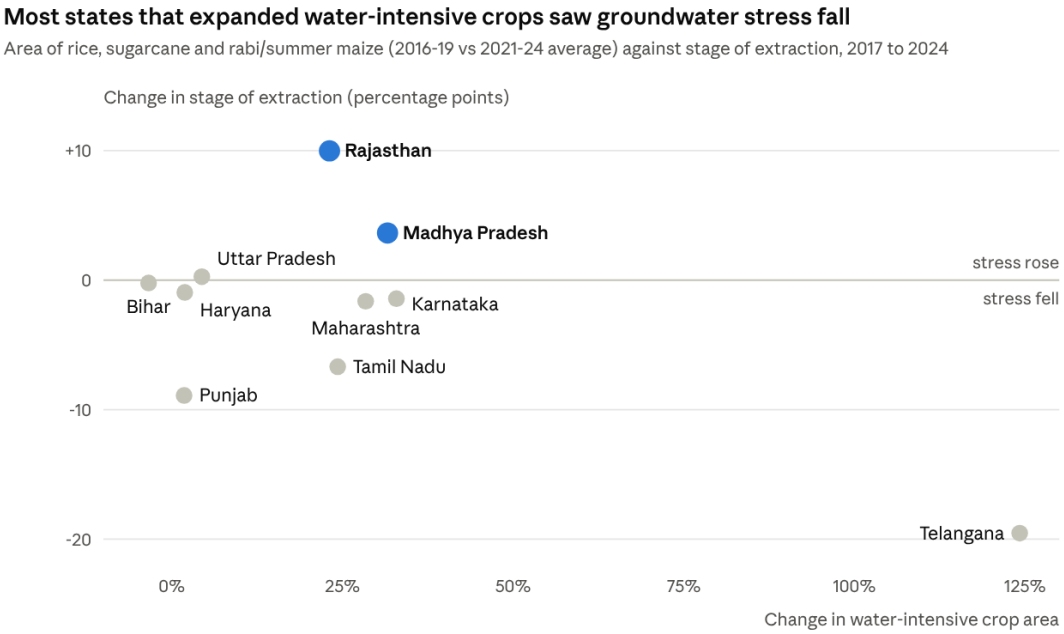


Figure 4 · UPAg (crop area); RBI Table 107 from CGWB (stage of extraction) · 10 states

State	Stage 2017	Stage 2024	Change (points)	Extractable resource change	Monsoon 2018–22 vs normal
Bihar	45.8%	45.5%	−0.2	+6.8%	−5%
Haryana	136.9%	136.0%	−0.9	+2.5%	−4%
Karnataka	69.9%	68.4%	−1.4	+14.1%	+15%
Madhya Pradesh	54.8%	58.4%	+3.6	−1.4%	+13%
Maharashtra	54.6%	53.0%	−1.6	+4.2%	+16%
Punjab	165.8%	156.9%	−8.9	−18.3%	−4%
Rajasthan	139.9%	149.9%	+10.0	−5.2%	+19%
Tamil Nadu	80.9%	74.3%	−6.7	+6.9%	+20%
Telangana	65.5%	45.9%	−19.5	+49.1%	+27%
Uttar Pradesh	70.2%	70.5%	+0.3	+1.6%	−14%

- **Expansion followed water.** The states that expanded most had the wettest monsoons and growing extractable groundwater; the extra recharge masks any extra pumping.

- **Correlations are weak.** Crop expansion vs change in stress: −0.22; vs change in extraction: +0.30; vs change in extractable resource: +0.52 (Spearman). None is significant with 10 states.
- **Two states break the pattern.** In Rajasthan and Madhya Pradesh stress rose while extractable groundwater fell, despite wet years.
- **Agriculture dominates extraction where we could check.** In Bihar, irrigation took 10.21 of 14.09 bcm in 2024 (72%) (CGWB Bihar report).

5.5 How much water sits behind ethanol

Why this step: the state test showed no link, so we checked how large ethanol's water use could be at most, to see whether it could matter at all.

Scenario	What it counts	2023-24 (bcm)	Share of extraction	2024-25 (bcm)	Share of extraction
S1: all crop water	Maize, rice and broken grain, cane	31.3–35.7	12.7–14.5%	50.0–60.1	20.4–24.4%
S3: by-products excluded	Maize, plus cane only if made from juice	13.4–23.2	5.4–9.4%	23.3–36.4	9.5–14.8%

These use total crop water per litre (maize 4,670 L, rice 10,790 L, cane 2,000–3,630 L) and the 2024 national extraction of 245.6 bcm. They are upper bounds for groundwater. National extraction fell from 248.7 bcm in 2017 to 245.6 bcm in 2024 while ethanol rose five-fold, so most of this water is rain, water already in use, or offset by wetter years. Section 6 therefore estimates only the new irrigation.

6. Cost translation

State statistics could not isolate an ethanol effect (Section 5.4), so we estimated from the bottom up only the new irrigation that the maize shift needs, and then put a price on it. The figures are gross: some new rabi maize replaced other rabi crops that also used water, and not all of it went to ethanol.

6.1 New groundwater from irrigated-season maize

Input	Low case	High case	Source
Added rabi + summer maize area	12.66 lakh ha (18.12 to 30.78)	same	UPAg, 2016-19 vs 2022-25 averages
Irrigation depth	300 mm (rabi maize: 5 irrigations of 6 cm)	700 mm (spring/summer maize: crop demand 693–955 mm)	Bharati et al. 2007, Bihar; Majumder et al. 2016, Punjab
Groundwater share of irrigation	61% (tube wells 45% + other wells 16% of net irrigated area)	100%	MoSPI Statistical Year Book, 2011-12

Result	Low case	High case
New irrigation water	3.8 bcm a year	8.9 bcm a year
New groundwater	2.3 bcm a year	8.9 bcm a year
Share of 2024 national extraction (245.6 bcm)	0.9%	3.6%

A cross-check on scale: maize output rose by 12.3 Mt between the 2016-19 and 2022-25 averages, which is about the maize used for ethanol in 2024-25 (12.5–13.9 Mt). The two match in size; that does not prove the extra maize went to ethanol, since feed demand and imports also moved.

6.2 Where it falls

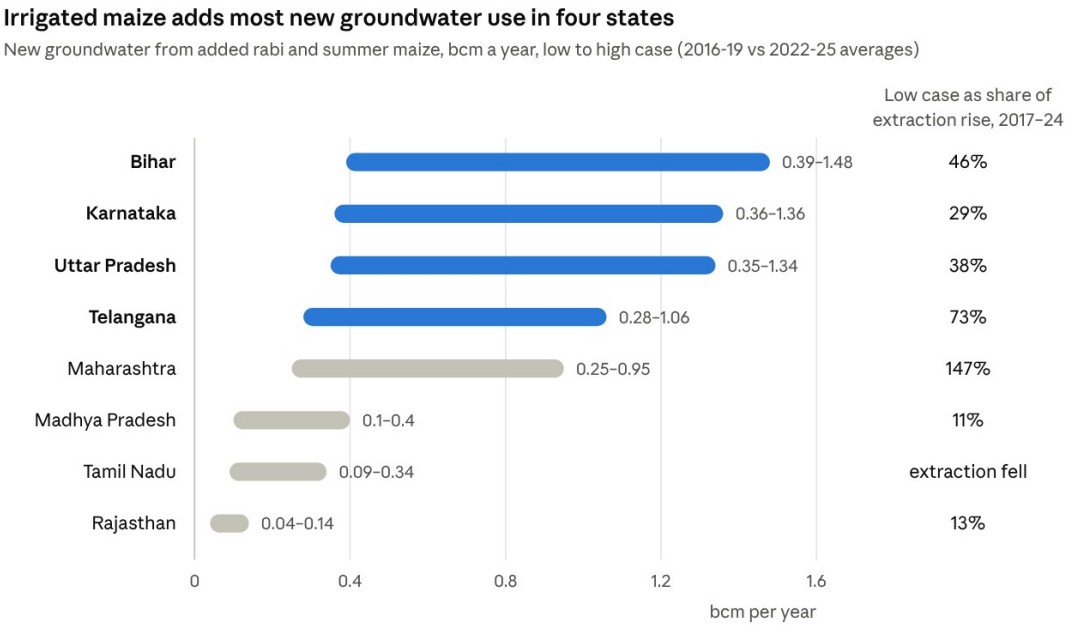


Figure 5 · Analysis_Stage4 (UPAg area; Bharati et al. 2007; Majumder et al. 2016; MoSPI; RBI Table 107) · 8 states with added rabi/summer maize

State	Added area (lakh ha)	New groundwater (bcm a year)	Extraction change 2017–24 (bcm)	Stage of extraction 2024
Bihar	2.11	0.39–1.48	+0.84	45.5% (safe)
Karnataka	1.94	0.36–1.36	+1.21	68.4% (safe, near 70%)
Uttar Pradesh	1.91	0.35–1.34	+0.92	70.5% (semi-critical)
Telangana	1.52	0.28–1.06	+0.38	45.9% (safe)
Maharashtra	1.36	0.25–0.95	+0.17	53.0% (safe)
Madhya Pradesh	0.57	0.10–0.40	+0.97	58.4% (safe)
Tamil Nadu	0.48	0.09–0.34	−0.28	74.3% (semi-critical)
Rajasthan	0.19	0.04–0.14	+0.28	149.9% (over-exploited)

- **Concentrated, not national.** These eight states hold about 80% of the added area. Bihar, Karnataka, Uttar Pradesh and Telangana alone account for about 60%.
- **It may matter where extraction did rise.** In Bihar, Karnataka, Uttar Pradesh and Telangana, even the low case equals roughly 30–75% of each state's rise in extraction since 2017, and in Maharashtra it exceeds the whole rise. This is suggestive only: we cannot net out crops that maize replaced.
- **State averages hide blocks.** "Safe" states such as Bihar and Telangana can still contain over-exploited blocks; we did not have block-level crop data to test this.

6.3 What a litre costs in water, money and carbon

Measure	Value	Basis
Groundwater per litre of irrigated-maize ethanol	about 860 L (rabi maize, low case) to 4,070 L (summer maize, high case)	depth ÷ yield ÷ 380–400 L/t × groundwater share

Measure	Value	Basis
Distillery process water, for comparison	3–5 L per litre	Rajya Sabha reply, Jul 2026
Farm power subsidy per m³ pumped (Punjab, 2020-21)	₹2.44	₹8,275 crore (Sonker & Kallummal, IIFT) ÷ 33.85 bcm (RBI Table 107)
Water subsidy cost per litre	₹2.1–9.95	2.9–13.8% of the ₹71.86 maize ethanol price (Lok Sabha AU755)
Subsidy on all new irrigated-maize groundwater	about ₹570–2,170 crore a year	new groundwater × ₹2.44 per m³
Forex saved per litre, ESY 2024-25	₹41.1	increase in cumulative savings (₹1,13,007 to ₹1,55,000 crore) ÷ 1,022.4 crore L (FIPI, Rajya Sabha via iamrenew, PPAC via BioEnergy Times)
CO_2 avoided per litre, ESY 2024-25	1.99 kg (about US$0.83 at a global social cost of carbon of $417/t)	official crude-substitution and CO_2 figures; Ricke et al. 2018

- **Nationally, the water cost looks smaller than the forex benefit.** Even the high case (₹9.95 per litre) is about a quarter of the forex saved per litre.
- **The subsidy is a floor, not the full cost.** It leaves out well deepening, energy on unsubsidised pumps and the value of aquifers that recharge slowly, so in over-exploited blocks the true cost could be much higher.
- **The CO_2 figure is the government's.** It likely leaves out fertiliser and land-use emissions, so we treat it as an upper bound on the benefit.
- **Process water is not the issue.** The groundwater behind a litre of irrigated-maize ethanol is roughly 170 to 1,400 times the distillery's own use, which is where current rules focus (Section 7, P2).

6.4 What we could not cost

- **Where each distillery buys its grain.** PPAC told the CIC it holds no supplier list (CIC order, Jul 2026), so feedstock cannot be traced to blocks.
- **Season-wise groundwater share for maize.** We used the national 2011-12 well share; a crop- and season-wise figure would narrow the 2.3–8.9 bcm range.
- **Displaced crops.** We have no data on what the new rabi maize replaced, which would turn gross water into net water.

7. Policy pathways

The findings point to the season and place of feedstock, not ethanol as such, as the main water risk. The options below follow from that. They are pathways worth testing, not tested recommendations: we have not modelled farmer response or the knock-on effects on food and feed prices.

P1. Season-aware feedstock pricing. Give a price premium or tender priority to ethanol from kharif (rain-fed) maize, damaged and surplus grain and crop residues (2G), with no premium for summer maize from over-exploited or critical blocks.

- *Evidence:* summer maize in 2024-25 covered 8.5 lakh ha and produced 38.5 lakh t, enough for about 146–154 crore L (14–15% of ESY 2024-25 blending), drawing an estimated 3.6–6.0 bcm of groundwater a year. Punjab already plans curbs on spring maize (The Tribune, Jun 2025).
- *Effect and trade-off:* avoids about ₹890–1,450 crore a year of power subsidy on that water. Cutting the volume outright would forgo about ₹6,000–6,300 crore of forex savings, so the

aim is to shift it to other feedstocks. Kharif grain needs storage, and some farmers would lose a third crop.

P2. Feedstock-aware distillery siting. When allotting ethanol tenders or interest subvention to new grain distilleries, consider the groundwater category of the blocks they source from, starting with Rajasthan, Punjab and Haryana (state stage above 135%).

- *Evidence:* the CGWA 2020 guidelines (via Drishti IAS) bar new industry in over-exploited units and require NOCs, but they cover only the plant's own 3–5 L per litre. Farmers are exempt, so the 860–4,070 L of irrigation groundwater per litre sits outside any rule.
- *Effect and trade-off:* targets the largest water term at little fiscal cost. It needs sourcing data (P5) and may raise transport costs.

P3. MSP with assured procurement for low-water crops in stressed states. Pair MSP for millets (jowar, bajra, ragi) with guaranteed procurement in arid and over-exploited states. This is the redistribution our original plan proposed.

- *Evidence:* millet MSP rose 106–160% but output grew only 8–39%, and the rank correlation of MSP rise with output growth is −0.43. Rice grew where procurement is assured. MSP alone did not shift crops; procurement may.
- *Effect and trade-off:* the size is uncertain; we found no state data to estimate the shift. Procurement and storage cost money, and the grain needs an outlet (public distribution, school meals).

P4. Micro-irrigation for rabi and summer maize and cane. Target drip and sprinkler subsidies at the irrigated-season crops that are growing.

- *Evidence:* micro-irrigation covers 115 lakh ha (8.1% of net sown area) and saves 20–48% of water (PIB, Aug 2026, citing the Economic Survey 2020-21).
- *Effect and trade-off:* applied to the new 2.3–8.9 bcm, about 0.5–4.3 bcm a year. We expect part of the saving to be lost if farmers use it to irrigate more land, so it works best alongside P1 or P2.

P5. Transparency. Publish feedstock-wise, state-wise ethanol supply, distillery water draw, and the season-wise groundwater share of irrigation by crop.

- *Evidence:* PPAC holds no supplier list (CIC order, Jul 2026), and the gaps in Sections 6.4 and 9 all trace to missing disclosure.
- *Effect and trade-off:* makes the ethanol–groundwater link testable at block level, at low cost.

How the pathways fit together. P5 comes first because it makes P1 and P2 targetable. P1 is the sharpest lever on our numbers, since summer maize is the thirstiest slice of ethanol supply grown specifically for it (surplus FCI rice and molasses exist anyway), and also the easiest to identify by season. P3 answers the MSP question we started with: our data supports using MSP to move farmers toward low-water crops only if it comes with procurement, as it does for rice and wheat.

8. Discussion and conclusions

We set out to test whether E20 is pushing India toward water-intensive crops and drawing down groundwater, and whether ethanol demand or MSP is the stronger driver. The evidence lets us say some things with moderate confidence, and others only as indications.

8.1 What the evidence suggests

- **E20's crop footprint is mainly maize, not sugarcane or rice.** Ethanol took roughly a fifth of India's maize in 2023-24 and close to a third in 2024-25. For rice (2–4%) and cane (0–10%) the share is small or depends on how the ethanol was made.

- **The extra maize is increasingly grown in the irrigated seasons.** Rabi and summer maize went from 27% to 43% of output. In Telangana and Uttar Pradesh, kharif maize output fell while rabi and summer output rose (Telangana 17.6 to 10.2 and 9.1 to 23.7 lakh t; Uttar Pradesh 13.6 to 6.6 and 1.6 to 13.0 lakh t, 2016-17 to 2024-25). That looks like a move into the irrigated season, not just extra land.
- **The groundwater effect is plausible but not yet visible in state statistics.** Our bottom-up estimate is 2.3–8.9 bcm a year of new groundwater (1–4% of national extraction). It is concentrated in Bihar, Karnataka, Uttar Pradesh and Telangana. Wetter monsoons in 2019–2022 raised recharge in most expanding states, which may be hiding the extra pumping.
- **Ethanol and MSP look less like rivals than partners.** Ethanol demand fits the maize boom. MSP fits rice only where it comes with procurement, and MSP alone did not move millets. Ethanol now also takes part of the procured rice surplus (23–24 lakh t of FCI rice in 2024-25; Deccan Herald, iamrenew). That gives MSP-driven paddy in water-stressed states a further outlet.
- **The water cost per litre appears modest next to the forex benefit, but it is a floor.** At ₹2.1–9.95 per litre against ₹41.1 of forex saved, the national trade-off looks favourable. The local cost in over-exploited blocks could be far higher than the subsidy suggests.

8.2 What we looked for, and what we found instead

Several of our original tests could not be run as planned because the data does not exist in public form. The table records what we did instead, so the reader can see where the conclusions are weaker.

We set out to show	What was missing	What we did instead, and found
Ethanol share of each crop, by state	Feedstock bought by each distillery (PPAC holds no supplier list)	National shares (maize 19–32%), with irrigated-season maize area as the state signal
Annual groundwater trend, 2015–2025	Annual stage of extraction (only 2017, 2020, 2022–2024 exist)	2017 vs 2024 with rainfall and recharge as controls: no significant link; wet monsoons masked pumping
Groundwater used by ethanol crops	Groundwater share of irrigation by crop, season and state	Bottom-up range from field irrigation depths and the 2011-12 well share: 2.3–8.9 bcm a year
Ethanol vs MSP as the cause	District or farm panel with prices and costs	Cross-crop comparison: MSP alone does not explain growth; ethanol fits maize; procurement fits rice
Other causes (industry, domestic)	Sector split of extraction by state	Bihar only (irrigation 72%); a distillery's own 3–5 L per litre is tiny beside its feedstock's irrigation
Cost of the water	Pumping cost and scarcity value	Punjab power subsidy per m^3: ₹2.1–9.95 per litre, a lower bound

8.3 Other findings worth noting

- **Official data may undercount irrigated maize in Punjab.** The Punjab Agriculture Department puts spring and summer maize at about 1.46 lakh ha and says it needs 18 or more irrigations (The Tribune, Jun 2025). UPAg shows no rabi or summer maize for Punjab, in the state with the highest stage of extraction.
- **Three cane and maize states sit near a category line.** Uttar Pradesh (70.5%) sits just above the 70% semi-critical mark, Karnataka (68.4%) is just below it, and Tamil Nadu (74.3%) is semi-critical. Small rises in extraction would change their category.
- **Rajasthan and Madhya Pradesh are the early warnings.** They are the only study states where stress rose and extractable groundwater fell despite wet years. Rajasthan has also seen protests against grain distilleries (Mongabay, Jul 2026).

- **The 30% target raises the stakes.** With blending now aimed at 30% by 2030 (iamrenew, Apr 2025), the irrigated-season trend matters more than the 2017–2024 averages.

8.4 Overall conclusion

Our reading is that E20's water risk is real but narrower than the original claim. It sits mainly in irrigated-season maize in a few states, it is small next to national extraction so far, and it does not yet show in state groundwater statistics, possibly because of wet years. Whether it becomes a depletion problem seems to depend on three things: how the 30% target is met, which feedstocks are used, and in which season the extra maize is grown. These conclusions rest on state averages and ranges, and could change with block-level and season-wise data.

9. Limitations and future work

Limitations.

- **Coverage.** Ten states and state averages; block-level hotspots can sit inside a stable state figure. With n = 10, none of the rank correlations is statistically significant.
- **Groundwater timing.** Only five assessment years exist, and the 2017 to 2024 comparison spans wet years that raised recharge.
- **Feedstock data.** The 2023-24 split comes from an Indian Express explainer and the 2024-25 split from an industry body (AIDA); neither is a primary government table.
- **Factors.** Conversion yields and water-per-litre figures come from press reports and an industry-commissioned study, and they are contested.
- **Irrigation inputs.** Irrigation depth comes from two field studies (Bihar and Punjab); the groundwater share of irrigation is a national 2011-12 figure.
- **Seasons.** Summer maize is reported separately only from 2022-23, and Punjab's spring maize appears to be missing from UPAg.
- **Gross, not net, water.** We could not subtract the water of crops that the new maize replaced.
- **Costs.** The Punjab power subsidy is used as a national proxy and is a lower bound. The CO_2 figure is the government's and probably leaves out fertiliser and land-use emissions.
- **Causation.** The study compares trends; it does not build a counterfactual. Rainfall data has a gap in 2017, and one 2021 Parliament answer had misaligned 2020 rows that we did not use.

Future work.

1. A block-level study of Bihar, Telangana, Rajasthan and Madhya Pradesh, matching CGWB block categories with district crop data.
2. Season-wise groundwater share of irrigation from the Minor Irrigation Census, to narrow the 2.3–8.9 bcm range.
3. Distillery sourcing data, through RTI requests or tender records, to trace feedstock to blocks.
4. Annual pre- and post-monsoon water levels from CGWB year books, as a yearly indicator between assessments.
5. Scenarios for 27% and 30% blending under different feedstock and season mixes.
6. Net water accounting that subtracts the crops the new maize replaced.

10. References

All pages were opened and checked on 25 October 2026. Each entry is also logged, with the figure it supports, in the dataset's Source_Log tab.

Official crop, price and groundwater data

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- DES Time Series, 3rd Advance Estimates 2025-26. All-India production.
- DES MSP statement, 12 Dec 2025 and CACP MSP compilation, 2012-13 to 2021-22.
- Sugarcane FRP: 2015-16, 2017-18, 2018-19 to 2022-23, 2024-25, 2025-26.
- RBI Handbook of Statistics on Indian States: Table 107, groundwater (2024-25 edition); 2023 edition Tables 60, 69, 71, 80.
- CGWB state reports: Punjab 2024, Bihar 2024; cross-check Lok Sabha AU3018.
- Sugarcane: ICAR-SBI state-wise APY 2018-19 to 2023-24, Lok Sabha AU1760, Lok Sabha AU443.
- Monsoon rainfall (MoES / IMD): LS 4446, Dec 2016, RS 165, Aug 2021, RS 25, Jul 2023.
- MoSPI Statistical Year Book, Irrigation chapter. Groundwater share of net irrigated area, 2011-12.

Ethanol policy, volumes, feedstock and prices

- National Policy on Biofuels 2018 (PM India); PRS on the 2018 MSP reset; PIB on advancing E20 (May 2022).
- NITI Aayog ethanol roadmap: PRS summary, full report.
- Parliament answers: Lok Sabha AU632 (distilleries), Lok Sabha AU2832 (blending, forex), Lok Sabha AU755 (maize ethanol price), Rajya Sabha reply, Jul 2026, Rajya Sabha reply, Dec 2025 (via iamrenew).
- Blending: MoPNG EBP page, Indian Chemical News (2020-21), Dataful dataset 21604, iamrenew, Apr 2025 (20% reached; 30% aim), PPAC via BioEnergy Times, Nov 2025.
- Feedstock split: Indian Express explainer via Vision IAS, Oct 2025, AIDA via Outlook Business, Feb 2026, Down To Earth, Aug 2025.
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Conversion factors, water factors and field studies

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- FIPI Policy & Economic Report, Jan 2025, p.51. Forex and crude substitution to ESY 2023-24.
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- CGWA 2020 groundwater guidelines (Drishti IAS summary).

Peer-reviewed literature

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- Dangar, S., Asoka, A. & Mishra, V. (2021). Journal of Hydrology. Repository record.
- Mishra, V. et al. (2024). Earth's Future. Repository record.

- Jha et al. (2026). PLOS ONE. Article (PLOS ONE).

Context

- The Hindu via Drishti IAS. India becomes a net maize importer.
- Mongabay India, Jul 2026. Water and food questions; Rajasthan distillery protests.

Appendix

A1. Dataset map (NeerNexus_Stage2_Dataset.xlsx, 17 tabs)

Tab	Contents
README	Purpose, conventions, tab guide
UPAg_crops	State × crop × season × year area and production, 2016-17 to 2025-26 (953 rows)
Groundwater	Extractable resource, extraction and stage of extraction, 10 states and All India, 2017–2024
Rainfall	Monsoon actual, normal and departure by state
Sugarcane, Rice, Coarse_cereals	Earlier state series from RBI, ICAR-SBI and Parliament answers
National_production	All-India production, 2015-16 to 2025-26
MSP_FRP	MSP and FRP, 2015-16 to 2025-26
Ethanol	Blending volumes, feedstock split, prices, distillery capacity
Source_Log	Every source with URL, original agency and type
Gaps	Data we could not find, and the workaround used
Analysis_H1 to Analysis_Water	Stage 3 tests (H1–H4, embedded water scenarios) with live formulas
Analysis_Stage4	New groundwater, cost per litre, state distribution, policy scenarios

A2. State maize production by season (UPAg, lakh tonnes)

State	Total 2016-17	Total 2024-25	Kharif 2016-17	Kharif 2024-25	Rabi + summer 2016-17	Rabi + summer 2024-25
Bihar	26.9	54.7	6.2	5.7	20.7	49.0
Haryana	0.2	0.2	0.2	0.2	0.0	0.0
Karnataka	33.1	61.9	29.9	49.6	3.2	12.2
Madhya Pradesh	33.4	66.4	32.6	57.9	0.8	8.5
Maharashtra	34.5	47.5	24.4	33.4	10.1	14.1
Punjab	4.5	3.7	4.5	3.7	0.0	0.0
Rajasthan	13.8	26.8	13.8	24.8	0.0	2.0
Tamil Nadu	9.5	26.6	7.7	13.8	1.8	12.8
Telangana	26.6	33.9	17.6	10.2	9.1	23.7
Uttar Pradesh	15.2	19.6	13.6	6.6	1.6	13.0
All India	259.0	434.1	189.2	248.1	69.8	186.0

A3. MSP rise and output growth, 2015-16 to 2024-25

Crop	MSP 2015-16 (₹/qtl)	MSP 2024-25 (₹/qtl)	MSP rise	Output 2015-16 (lakh t)	Output 2024-25 (lakh t)	Output growth
Paddy / rice	1,410	2,300	+63.1%	1,044.1	1,501.8	+43.8%
Wheat	1,525	2,425	+59.0%	922.9	1,179.5	+27.8%

Crop	MSP 2015-16 (₹/qtl)	MSP 2024-25 (₹/qtl)	MSP rise	Output 2015-16 (lakh t)	Output 2024-25 (lakh t)	Output growth
Maize	1,325	2,225	+67.9%	225.7	434.1	+92.4%
Jowar (hybrid)	1,570	3,371	+114.7%	42.4	49.6	+16.9%
Bajra	1,275	2,625	+105.9%	80.7	112.1	+38.9%
Ragi	1,650	4,290	+160.0%	18.2	19.8	+8.5%
Sugarcane (FRP)	230	340	+47.8%	3,484.5	4,546.1	+30.5%

Wheat MSP is listed by crop year (2024-25 = RMS 2025-26). The FRP base recovery rose from 9.5% to 10.25%; at 9.5% recovery the 2024-25 FRP is ₹315.10, a 37% rise.

A4. Stage of groundwater extraction (CGWB via RBI Table 107)

State	2017	2020	2022	2023	2024
Bihar	45.8%	51.1%	44.9%	44.8%	45.5%
Haryana	136.9%	134.6%	134.1%	135.7%	136.0%
Karnataka	69.9%	64.9%	69.9%	66.3%	68.4%
Madhya Pradesh	54.8%	56.8%	59.1%	58.8%	58.4%
Maharashtra	54.6%	55.0%	54.7%	53.8%	53.0%
Punjab	165.8%	164.4%	164.1%	163.8%	156.9%
Rajasthan	139.9%	150.2%	151.1%	148.8%	149.9%
Tamil Nadu	80.9%	82.9%	75.6%	73.9%	74.3%
Telangana	65.5%	53.3%	41.6%	38.7%	45.9%
Uttar Pradesh	70.2%	68.8%	70.7%	70.8%	70.5%
All India	63.3%	61.6%	60.1%	59.3%	60.5%

Punjab 2022 uses the CGWB Punjab report (164.11%); RBI prints 165.99%.

A5. Monsoon rainfall departure from normal (MoES / IMD)

State	2015	2016	2018	2019	2020	2021	2022	Average 2018–22
Bihar	−27%	−3%	−25%	+3%	+25%	+3%	−31%	−5%
Haryana	−37%	−28%	−5%	−42%	−14%	+30%	+9%	−4%
Karnataka	−20%	−15%	−6%	+23%	+27%	+3%	+29%	+15%
Madhya Pradesh	−11%	+19%	−8%	+44%	+5%	0%	+23%	+13%
Maharashtra	−27%	+16%	−8%	+32%	+16%	+19%	+23%	+16%
Punjab	−31%	−28%	+16%	−7%	−17%	−7%	−6%	−4%
Rajasthan	+10%	+28%	−6%	+41%	+8%	+17%	+37%	+19%
Tamil Nadu	−10%	−19%	−8%	+17%	+27%	+17%	+45%	+20%
Telangana	−20%	+19%	−3%	+6%	+46%	+39%	+46%	+27%
Uttar Pradesh	−46%	−14%	−10%	−9%	−18%	−5%	−29%	−14%

No 2017 figures were found in the Parliament answers used.

A6. Key formulas

- Feedstock share: ethanol (L) ÷ yield (L/t) ÷ national production (t).
- New groundwater: added rabi + summer maize area × irrigation depth × groundwater share.
- Groundwater per litre: irrigation depth × 10 ÷ yield (t/ha) ÷ ethanol yield (L/t) × groundwater share, in m³ per litre.
- Water subsidy per litre: groundwater per litre × Punjab subsidy per m³ (₹8,275 crore ÷ 33.85 bcm = ₹2.44).

- Forex per litre (ESY 2024-25): (₹1,55,000 − ₹1,13,007 crore) ÷ 1,022.4 crore L = ₹41.1.
- CO_2 per litre: (260 − 193 lakh t crude) ÷ 1,022.4 crore L × (790 ÷ 260 t CO_2 per t crude) = 1.99 kg.

Groundwater per litre of ethanol = (irrigation depth × 10) ÷ (crop yield × ethanol yield per tonne) × groundwater share of irrigation